

Screen Clean: Safe and Gap-Safe Rules for LASSO in High Dimensions

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1 Introduction

Modern data sets often contain a large number of candidate predictors while only a few truly influence the response. Eliminating the irrelevant features is therefore essential. This problematic belongs to the *compressive sensing* field which seeks solutions with sparse¹ support. A common approach is the Least Absolute Shrinkage and Selection Operator (LASSO), defined below, which drives some coefficients exactly to zero and thereby performs variable selection. Nonetheless, on very high dimension problem, it can be very time consuming and memory intensive to use standard methods.

A powerful solution is *safe screening*. Before (or during) optimization, it identifies variables whose optimal LASSO coefficients are guaranteed to be zero and discards them. The test is exact, no relevant variable is ever removed, the reduced problem is strictly equivalent to the original one, while its dimension is often orders of magnitude smaller [GVR10]. So, screening accelerates existing solvers, lowers memory requirements and extends sparse learning to problem sizes that would otherwise be out of reach [FGS15].

In this paper we review screening strategies, explain the convex analytic principles that enable them, and quantify the computational gains they deliver with experimental results.

1.1 LASSO

In the suite of the report, we let $m \in \mathbb{N}^*$ the number of measurements and $n \in \mathbb{N}^*$ the number of features. So given $y \in \mathbb{R}^m$ and a design matrix $A = [a_1, \dots, a_m] \in \mathbb{R}^{n \times m}$, LASSO introduced by [Tib96], is the convex program

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} \|y - Ax\|_2^2 + \lambda \|x\|_1, \quad \lambda > 0 \quad (\text{LASSO})$$

The objective is coercive and strictly convex so the problem has a unique minimizer, denote by x_λ^* .

The ℓ_1 penalty allows to shrink irrelevant features to zero according to the choice of λ . There exists a threshold where above it, all coefficients are zero.

Proposition 1. *For every $\lambda \geq \lambda_0 := \|A^T y\|_\infty$ the unique solution of the (LASSO) is identically zero.*

Proof. We have,

$$\frac{1}{2} \|y - Ax\|_2^2 = \frac{1}{2} \|y\|_2^2 - \langle Ax, y \rangle + \frac{1}{2} \|Ax\|_2^2 \geq \frac{1}{2} \|y\|_2^2 - |\langle Ax, y \rangle|$$

and using the Hölder's inequality

$$|\langle Ax, y \rangle| = |\langle x, A^T y \rangle| \leq \|x\|_1 \|A^T y\|_\infty$$

So, if $\lambda \geq \|A^T y\|_\infty$,

$$\forall x \in \mathbb{R}^n, \quad \frac{1}{2} \|y - Ax\|_2^2 + \lambda \|x\|_1 \geq \frac{1}{2} \|y\|_2^2 + (\lambda - \|A^T y\|_\infty) \|x\|_1 \geq \frac{1}{2} \|y - 0\|_2^2 + 0$$

Then (LASSO) has its minimizer null. □

We will state some results and reformulate (LASSO) to explain the key idea behind screening.

¹vectors containing only a few non-zero coefficients

1.2 Primal–dual problem

Definition 1 (Convex optimization problem). *Let $n, p, q \in \mathbb{N}^*$ a problem of the form*

$$\begin{aligned} \min_{x \in \mathbb{R}^n} & f(x) \\ \text{s.t. } & g_i(x) \leq 0 \quad i \in \llbracket 1 \dots p \rrbracket \\ & h_j(x) = 0 \quad j \in \llbracket 1 \dots q \rrbracket \end{aligned}$$

is called a convex optimization problem if f and each g_i are convex on $\mathbb{R}^n$ and each h_j is linear.

We have that (LASSO) is a convex optimization problem for $\lambda > 0$ because the objectif is convex and has no constraint. Moreover, if we let $z := Ax$, to introduce a constraint, we obtain a *primal problem*,

$$(\mathcal{P}_\lambda) : \min_{(x,z) \in \mathbb{R}^n \times \mathbb{R}^m} P_\lambda((x,z)) := \frac{1}{2} \|y - z\|_2^2 + \lambda \|x\|_1, \quad \text{s.t. } z = Ax \quad (1)$$

because all decision variables (here the coefficients of x and z) appear explicitly in the objective and in the constraints.

We form the Lagrangian ²,

$$\mathcal{L} : ((x, z), u) \in \mathbb{R}^{n+q} \times \mathbb{R}^m \mapsto \frac{1}{2} \|y - z\|_2^2 + \lambda \|x\|_1 + \langle u, z - Ax \rangle$$

We call *dual function*,

$$D_\lambda : u \in \mathbb{R}^m \mapsto \inf_{(x,z) \in \mathbb{R}^n \times \mathbb{R}^m} \mathcal{L}((x,z), u)$$

And we define a *dual problem*,

$$(\mathcal{D}_\lambda) : \max_{u \in \mathbb{R}^m} D_\lambda(u) \quad (2)$$

We simplify the expression of D_λ .

Proposition 2. *For $\lambda > 0$, $\mathcal{P}_\lambda$ admits as dual problem,*

$$(\mathcal{D}_\lambda) : \max_{u \in \mathbb{R}^m} -\frac{1}{2} \|u\|_2^2 + \langle u, y \rangle \quad \text{s.t. } \|A^T u\|_\infty \leq \lambda \quad (3)$$

Proof. We can split the infimum of D_λ according to primal variables. We minimizing according to z . So we fixed x, u and we remark $z \mapsto \mathcal{L}((x, z), u)$ is strictly convex and coercive so it admits a unique solution, z^* , and verify,

$$\partial_z \mathcal{L}((x, z^*), u) = 0 \Rightarrow -(y - z^*) + u = 0 \Rightarrow z^* = y - u$$

so, the minimal value is,

$$\inf_{z \in \mathbb{R}^m} \mathcal{L}((x, z), u) = \frac{1}{2} \|u\|_2^2 + \lambda \|x\|_1 + \langle u, y - u - Ax \rangle = -\frac{1}{2} \|u\|_2^2 + \langle u, y \rangle + \lambda \|x\|_1 + \langle A^T u, x \rangle$$

We now minimize over x . Since the first two terms don't depend on x , we act on each coordinate of x separately,

$$\inf_{x \in \mathbb{R}^n} (\lambda \|x\|_1 + \langle A^T u, x \rangle) = \sum_{i=1}^n \inf_{t \in \mathbb{R}} (\lambda |t| + [A^T u]_i t)$$

Let $i \in \llbracket 1, n \rrbracket$, we have,

$$|[A^T u]_i| \leq \lambda \implies \lambda |t| + [A^T u]_i t \geq \lambda |t| - |[A^T u]_i| |t| \geq 0$$

and if $|[A^T u]_i| > \lambda$, then by choosing $t \rightarrow \pm\infty$ with the same sign as $[A^T u]_i$,

$$\lambda |t| + [A^T u]_i t \longrightarrow -\infty$$

²There is no inequality constraint, so we have two and not three arguments.

Hence,

$$\inf_{x \in \mathbb{R}^n} \left(\lambda \|x\|_1 + \langle A^T u, x \rangle \right) = \begin{cases} \sum_{i=1}^n 0 & \forall i \in \llbracket 1, n \rrbracket, |[A^T u]_i| \leq \lambda \\ -\infty & \text{otherwise} \end{cases} = \begin{cases} 0 & \|A^T u\|_\infty \leq \lambda \\ -\infty & \text{otherwise} \end{cases}$$

We remark that the feasible dual set constraints the space. Consequently, we have

$$(\mathcal{D}_\lambda) : \max_{u \in \mathbb{R}^m} -\frac{1}{2} \|u\|_2^2 + \langle u, y \rangle \quad \text{s.t.} \quad \|A^T u\|_\infty \leq \lambda \quad (4)$$

□

Remark 1. Note that (4) is equivalent to

$$(\mathcal{D}_\lambda) : \max_{u \in \mathbb{R}^m} \frac{1}{2} \|y\|_2^2 - \frac{1}{2} \|y - u\|_2^2 \quad \text{s.t.} \quad \|A^T u\|_\infty \leq \lambda \quad (5)$$

To show that, we develop (5):

$$\begin{aligned} \frac{1}{2} \|y\|_2^2 - \frac{1}{2} \|y - u\|_2^2 &= \frac{1}{2} \|y\|_2^2 - \frac{1}{2} (\|y\|_2^2 - 2\langle y, u \rangle + \|u\|_2^2) \\ &= \frac{1}{2} \|y\|_2^2 - \frac{1}{2} \|y\|_2^2 + \langle y, u \rangle - \frac{1}{2} \|u\|_2^2 \\ &= -\frac{1}{2} \|u\|_2^2 + \langle u, y \rangle. \end{aligned}$$

In the following, we use those two formulation of $(\mathcal{D}_\lambda)$ identically. We also note that some authors (e.g., [GVR10]) use the formulation

$$(\mathcal{D}_\lambda) : \max_{u \in \mathbb{R}^m} \frac{1}{2} \|y\|_2^2 - \frac{1}{2} \|y + u\|_2^2 \quad \text{s.t.} \quad \|A^\top u\|_\infty \leq \lambda,$$

which is identical to (5) up to a sign change. This difference arises because their Lagrangian is chosen as

$$\mathcal{L}(x, z, u) = \frac{1}{2} \|y - z\|_2^2 + \lambda \|x\|_1 + \langle u, Ax - z \rangle.$$

Definition 2 (Slater's condition [Lee21]). Let $\mathcal{P}$ a convex optimization problem, with the same notation as Definition 1. We say that Slater's condition holds if it exists a point $x \in \mathbb{R}^n$ such that,

$$g_i(x) < 0 \quad i \in \llbracket 1, p \rrbracket \quad h_j(x) = 0 \quad j \in \llbracket 1, q \rrbracket$$

Remark 2. For every $\lambda > 0$, $\mathcal{P}_\lambda$ has only a linear equality constraint. For any $x \in \mathbb{R}^n$, (x, Ax) satisfies this constraint so Slater's condition holds for $\mathcal{P}_\lambda$

1.3 Screening for the LASSO

Before to see how to use the dual problem to do screening, we remind a famous theorem in optimization. We need the introduction of *sub-differential* for convex function, which coincides with the usual derivative when the function is differentiable.

Definition 3 (Subdifferential). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function and $x \in \mathbb{R}^n$. The subdifferential of f at x is the set of all subgradient,

$$\partial f(x) = \{ g \in \mathbb{R}^n, \forall h \in \mathbb{R}^n \quad f(x+h) \geq f(x) + \langle g, h \rangle \}$$

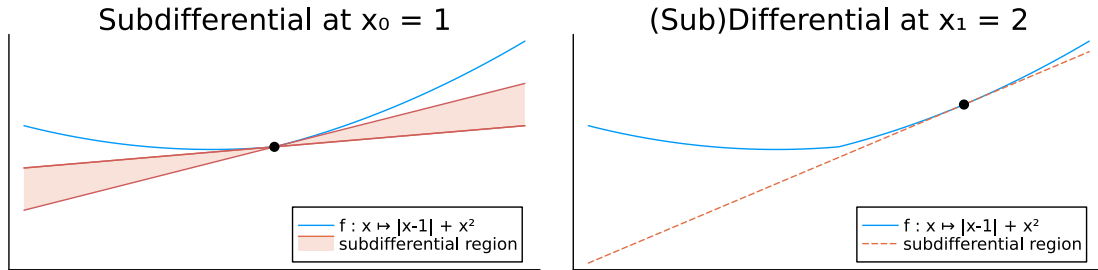


Figure 1: Let $f : x \in \mathbb{R} \mapsto |x-1| + x^2$, be differentiable on $\mathbb{R} \setminus \{1\}$. At $x_0 = 1$, (left) $\partial f(x_0) = [1, 3]$ is the set of slopes passing through x_0 and under f . Then, at $x_1 = 2$ (right), $\partial f(x_1) = \{5\} = \{f'(2)\}$, so the subdifferential coincides to its unique tangent.

Theorem 1 (Karush–Kuhn–Tucker, sufficient and necessary conditions [BV04, Chapter 5.]). *Let a convex optimization problem as in Definition 1, and assume Slater's condition holds. Then a primal–dual triple*

$$(w^*, \mu^*, u^*) \in \mathbb{R}^n \times \mathbb{R}_+^m \times \mathbb{R}^p$$

*is optimal*³ *if and only if it satisfies*

$$\left\{ \begin{array}{ll} \text{(i) Primal feasibility:} & g_i(w^*) \leq 0 \quad i \in \llbracket 1, p \rrbracket \\ & h_j(w^*) = 0 \quad j \in \llbracket 1, q \rrbracket \\ \text{(ii) Dual feasibility:} & \mu_i^* \geq 0 \quad i \in \llbracket 1, p \rrbracket \\ \text{(iii) Complementary slackness:} & \mu_i^* g_i(w^*) = 0 \quad i \in \llbracket 1, p \rrbracket \\ \text{(iv) Stationarity:} & 0 \in \partial f(w^*) + \sum_{i=1}^p \mu_i^* \partial g_i(w^*) + \sum_{j=1}^q \partial h_j(w^*) u_j^* \end{array} \right.$$

We are now able to get a test to do screening on (LASSO).

Proposition 3 (Safe screening via KKT). *For $\lambda > 0$, Let x_λ^* the unique minimizer of (LASSO). We set $u_\lambda^* := y - Ax_\lambda^*$, then u_λ^* is optimal for $\mathcal{D}_\lambda$ and for each feature $i \in \llbracket 1, n \rrbracket$,*

$$|a_i^T u_\lambda^*| < \lambda \implies (x_\lambda^*)_i = 0$$

Proof. Let (x^*, z^*) solving $\mathcal{P}_\lambda$ and u^* solving $\mathcal{D}_\lambda$. We have according to $\mathcal{P}_\lambda$, $z^* = Ax^*$

$((x^*, z^*), \mathbf{0}, u^*)$ is an optimal primal-dual triple so we have from Stationarity of Theorem 1, by taking the subdifferential according to $\begin{pmatrix} x \\ z \end{pmatrix}$ and separate according to the dependance of each variable,

$$0 \in \partial_x \left(\lambda \|x\|_1 - \langle A^T u^*, x \rangle \right) (x^*) \quad \text{and} \quad 0 \in \partial_z \left(\frac{1}{2} \|y - z\|_2^2 + \langle u^*, z \rangle \right) (z^*)$$

Using the second condition, we get, by differentiability,

$$-(y - z^*) + u^* = 0 \iff u^* = y - z^*$$

Using the first condition, we get,

$$A^T u^* \in \lambda \partial_x (\|x\|_1) (x^*)$$

We remind, for $x = (x_1, \dots, x_n) \in \mathbb{R}^n$,

$$\partial \|\cdot\|_1(x) = \partial |\cdot|(x_1) \times \dots \times \partial |\cdot|(x_n) \quad \text{and} \quad \partial |\cdot|(x_1) = \begin{cases} \{\text{sign}(x_1)\} & \text{if } x_1 \neq 0 \\ [-1, 1] & \text{if } x_1 = 0 \end{cases}$$

Thus,

$$\forall i \in \llbracket 1, n \rrbracket, \quad a_i^T u^* \in \begin{cases} \{\lambda \text{sign}(x_i^*)\} & \text{if } x_i^* \neq 0 \\ [-\lambda, \lambda] & \text{if } x_i^* = 0 \end{cases}$$

So whenever $|a_i^T u^*| < \lambda$, the only choice is $x_i^* = 0$. This is the screening rule. $\square$

Next, we remark the usefulness of the dual for many reasons. We have a problem with constraint, helping the optimization in a smaller space than the primal. Indeed, its a typical case in variable selection to have $m \ll n$. More than just reducing dimension, the dual yields a powerful screening rule.

³(x^* solves the primal and (μ^*, u^*) solves the dual)

2 Safe

We can't use directly the Proposition 3 because u_λ^* is unknown. So, we will work on sets where we are sure that contain u_λ^* .

Definition 4 (Safe region). *For $\lambda > 0$, we call $\mathcal{R} \subset \mathbb{R}^m$ a safe region of $(\mathcal{D}_\lambda)$ if $u_\lambda^* \in \mathcal{R}$*

Definition 5 (Safe rule, [GVR10]). *For $\lambda > 0$, any safe region $\mathcal{R}$, $\forall i \in \llbracket 1, n \rrbracket$ and any primal optimal solution x_λ^* , we call safe rule, a test associated to $\mathcal{R}$, $\mu_{\mathcal{R}}$, if the following holds true,*

$$\mu_{\mathcal{R}}(a_i) = \max_{u \in \mathcal{R}} |a_i^T u| < \lambda \implies (x_\lambda^*)_i = 0$$

A natural goal is to find safe regions as small as possible: narrowing safe region can increase the number of screened out variables and consequently decrease the memory cost (because there are few variable to stock) which at the end accelerate the optimization solver.

In the following sections, we are going to construct safe regions and derive the corresponding safe rule. We seek to apply Safe for $\mathcal{P}_\lambda$ with $\lambda < \lambda_0$, where λ_0 is the smallest value to be large enough so that $x_{\lambda_0}^*$ is identically zero, that is $\lambda_0 = \|A^T y\|_\infty$, see Proposition 1. In this case $u_0^* = y$ (see proposition 3)

2.1 Construction of Safe regions

2.1.1 Safe region $\mathcal{R}_1$: Ball

We seek to construct a safe region that is both convex and easy to manipulate. Naturally, we therefore choose to form a ball.

Proposition 4. *Let $0 < \lambda < \lambda_0$ and u_0^* the optimal solution of $\mathcal{D}_{\lambda_0}$. The ball $\mathcal{B}(c, r)$ where center $c = y$ and the radius $r = \lambda \|y\|_2 |1/\lambda - 1/\lambda_0|$ is a safe region for u_0^* .*

Proof. Let u in $\mathcal{B}(y, r)$,

$$\|u - y\|_2^2 \leq r^2 = \lambda^2 \|y\|_2^2 \left| \frac{1}{\lambda} - \frac{1}{\lambda_0} \right|^2 = \|y - \frac{\lambda}{\lambda_0} y\|_2^2$$

Let $u_\rho = \frac{\lambda}{\lambda_0} u_0^*$ be a dual feasible point of $(\mathcal{D}_\lambda)$. Indeed, we have,

$$\|A^T u_\rho\| = \frac{\lambda}{\lambda_0} \|A^T u_0^*\| < \frac{\lambda}{\lambda_0} \lambda = \lambda$$

The previous expression becomes

$$\|u - y\|_2^2 \leq \|y - u_\rho\|_2^2$$

$$\frac{1}{2} \|y\|_2^2 - \frac{1}{2} \|y - u\|_2^2 \geq \frac{1}{2} \|y\|_2^2 - \frac{1}{2} \|y - u_\rho\|_2^2$$

That means

$$D_\lambda(u) \geq D_\lambda(u_\rho)$$

Hence,

$$\mathcal{B}(y, r) = \{u \mid D_\lambda(u) \geq D_\lambda(u_\rho)\}$$

The optimality condition of $\mathcal{D}_\lambda$ states that $D_\lambda(u^*) \geq D_\lambda(u)$ for all u in the feasible set of $(\mathcal{D}_\lambda)$. In particular, we have $D_\lambda(u_\lambda^*) \geq D_\lambda(u_\rho)$, so $u_\lambda^* \in \mathcal{B}(c, r)$. Therefore $\mathcal{B}(c, r)$ is indeed a safe region of $(\mathcal{D}_\lambda)$. $\square$

The choice of u_ρ in the proof is motivated by the fact that linear scaling allows us to manipulate the feasible set more easily and, at the same time, produces the smallest safe region.

2.1.2 Safe region $\mathcal{R}_2$: Half plane

Proposition 5. *Let $g = \partial D_{\lambda_0}(u_0^*) = u_0^* - y$. The set $\mathcal{R}_2 = \{u \in \mathbb{R}^m, g^T(u - u_0^*) \leq 0\}$ is a safe region.*

Proof. Because $0 < \lambda < \lambda_0$, any u feasible for $(\mathcal{D}_\lambda)$ also satisfies $\|A^T u\|_\infty \leq \lambda_0$, and is therefore feasible for $(\mathcal{D}_{\lambda_0})$. By the first-order (subgradient) optimality condition for the concave maximization defining $\mathcal{D}_{\lambda_0}$, we have

$$g^T(u - u_0^*) \leq 0 \quad \text{for all feasible } u.$$

Taking $u = u_\lambda^*$ gives

$$g^T(u_\lambda^* - u_0^*) \leq 0,$$

so $\mathcal{R}_2$ is a safe region of $(\mathcal{D}_\lambda)$. □

Remark 3. $\mathcal{R}_2$ is a closed half-space whose boundary is the tangent hyperplane at u_0^* . In fact,

$$\mathcal{R}_2 = \{u \in \mathbb{R}^m \mid g^\top(u - u_0^*) \leq 0\}$$

is the zero-sublevel set ⁴ of the affine functional $u \mapsto g^\top(u - u_0^*)$. Since

$$u \mapsto g^\top(u - u_0^*)$$

is continuous and linear, its zero-sublevel set is closed and convex, and in fact is a (closed) half-space.

Its boundary is the zero-level set

$$H = \{u \mid g^\top(u - u_0^*) = 0\},$$

which is an affine hyperplane passing through u_0^* with normal vector g .

Finally, because $\partial D_{\lambda_0}(u_0^*) = g$, the hyperplane H is exactly the tangent hyperplane to the level set $\{D_{\lambda_0}(u) = D_{\lambda_0}(u_0^*)\}$ at the point u_0^* ⁵. Hence $\mathcal{R}_2$ is the closed half-space bounded by that tangent hyperplane.

2.1.3 Safe region $\mathcal{R}$: Hemisphere

By intersecting these two precedents regions, $\mathcal{R}_1, \mathcal{R}_2$, we obtain an even tighter safe region, $\mathcal{R} := \mathcal{R}_1 \cap \mathcal{R}_2$. It allows for more aggressive variable screening. The intersection of a half-plane by a ball is named a dome and by the Remark 3 it becomes a particular dome, a hemisphere.

⁴For a function $f : X \rightarrow \mathbb{R}$ and a scalar $\alpha \in \mathbb{R}$, the α -sublevel set is defined by

$$L_\alpha(f) = \{x \in X : f(x) \leq \alpha\}.$$

⁵Indeed, for any differentiable function f , the tangent hyperplane to the level set $f(x) = f(y)$ is given by

$$\{x \mid \partial f(y)^\top(x - y) = 0\}$$

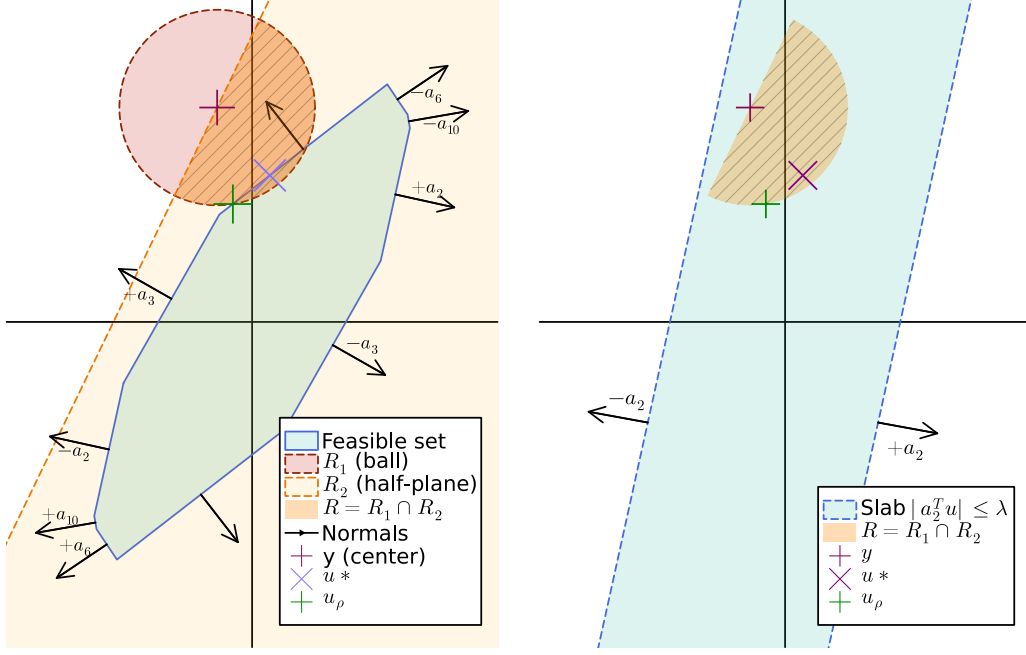


Figure 2: Dual-space view of Safe for (LASSO) on a toy instance ($m = 2$, $n = 10$, $\lambda = \rho\lambda_0$). *Left*: the feasible polytope $\mathcal{F}_\rho = \{u : \|A^\top u\|_\infty \leq \lambda\}$, the Safe ball $\mathcal{R}_1$, the half-space $\mathcal{R}_2$ (boundary through u_0^* with normal $u_0^* + y$) and their intersection $\mathcal{R} = \mathcal{R}_1 \cap \mathcal{R}_2$ (hatched). Outward normals of the active constraints are $\pm a_i$. *Right*: slab $S_2 = \{u \in \mathbb{R}^m : |a_2^\top u| \leq \lambda\}$. Because $R \subset \mathring{S}_2$ we have $|a_2^\top u^*| < \lambda$ and by the safe rule we conclude $x_2^* = 0$.

We remark on Figure 2.1.3 that if we consider only $\mathcal{R}_1$ or $\mathcal{R}_2$ we can't know if x_2 is zero because at each time we would have $R \not\subset \mathring{S}_2$. So narrowing it is useful.

In what follows, we derive the Safe rule corresponding to this refined region.

2.2 Explicit Safe screening rule

Now, we have built a Safe region $\mathcal{R} = \mathcal{R}_1 \cap \mathcal{R}_2$, we can then derive the Safe rule corresponding to it.

Theorem 2 (Safe Lasso rule, [GVR10]). *Let $A \in \mathbb{R}^{m \times n}$, $y \in \mathbb{R}^m$, and for each $\lambda > 0$, let*

$$u_\lambda^* = \min_u \|u - y\|_2, |A^\top u|_\infty \leq \lambda$$

be the dual optimal solution of LASSO at λ . Assume that $u_{\lambda_0}^$ is known for some $\lambda_0 > 0$. Then for any $\lambda > 0$, the i -th primal coefficient, the corresponding safe rule is:*

$$\mu_{\mathcal{R}_1 \cap \mathcal{R}_2}(a_i) = |a_i^\top u_{\lambda_0}^*| + \lambda \|a_i\|_2 \|y\|_2 \left| \frac{1}{\lambda} - \frac{1}{\lambda_0} \right|$$

Proof. By proposition 4, the new dual optimum u_λ^* lies in $\mathcal{R} = \mathcal{R}_1 \cap \mathcal{R}_2$ and hence lies in $\mathcal{R}_1$

$$u_\lambda^* \in B(u_{\lambda_0}^*, \rho), \quad \rho = \lambda \|y\|_2 \left| \frac{1}{\lambda} - \frac{1}{\lambda_0} \right|.$$

Thus every feasible dual u at λ can be written $u = u_{\lambda_0}^* + v$, with $\|v\|_2 \leq \rho$. Hence for any feature a_i ,

$$\max_{u \in \mathcal{R}} |a_i^\top u| = \max_{\|v\| \leq \rho} |a_i^\top (u_{\lambda_0}^* + v)| = |a_i^\top u_{\lambda_0}^*| + \|a_i\|_2 \rho$$

□

Theorem 3. *The Safe rule from Theorem 2 is useless as soon as $\frac{\lambda}{\lambda_0} \leq \min_{j \in \llbracket 1, m \rrbracket} \left(\frac{1 + |a_j^T y| / (\|a_j\| \|y\|)}{1 + \lambda_0 / (\|a_j\| \|y\|)} \right)$*

Proof. Recall that the screening test for feature j discards j if

$$\max_{u \in \mathcal{R}} |a_j^T u| < \lambda, \text{ where } \mathcal{R} = \mathcal{R}_1 \cap \mathcal{R}_2$$

By Theorem 2,

$$\max_{u \in \mathcal{R}} |a_j^T u| = |a_j^T y| + r \|a_j\| = \|a_j\| \|y\| \left(\frac{|a_j^T y|}{\|a_j\| \|y\|} + 1 - \frac{\lambda}{\lambda_0} \right)$$

The test never discards feature j exactly when

$$|a_j^T y| + r \|a_j\| \geq \lambda$$

i.e.

$$\|a_j\| \|y\| \left(1 + \frac{|a_j^T y|}{\|a_j\| \|y\|} - \frac{\lambda}{\lambda_0} \right) \geq \lambda$$

Rearranging yields

$$1 + \frac{|a_j^T y|}{\|a_j\| \|y\|} \geq \frac{\lambda}{\lambda_0} + \frac{\lambda}{\|a_j\| \|y\|} \iff \frac{\lambda}{\lambda_0} \leq \frac{1 + |a_j^T y| / (\|a_j\| \|y\|)}{1 + \lambda_0 / (\|a_j\| \|y\|)}$$

Thus if λ/λ_0 is at most the minimum of these right-hand sides over j , then for every j , $\max_{u \in \mathcal{R}} |a_j^T u| \geq \lambda$, so no feature can be eliminated. This shows the rule is useless under the stated condition. $\square$

3 Gap Safe

Safe Lasso screening rules discard variables once, before the optimisation begins. We called this kind of safe rule, *static safe rule* because during the iterations no fresh variables can be eliminated.

In general, first-order solvers for convex problems produce an iterative sequence $(x_k)_{k \geq 0}$ that converges to the unique primal optimum x^* while monotonically decreasing the objective value, like ISTA [BT09], Coordinate Descent [FHT10], ... So to benefit the increasingly accurate information got at each iteration we will construct a collection of nested regions refereed as *dynamic*.

Definition 6 (Convergent collection of safe regions). *Let $(\mathcal{R}_k)_{k \geq 0}$ be a sequence of safe regions for u^* , the optimal of the dual problem. We say that $(\mathcal{R}_k)$ is convergent if,*

$$\text{diam}(\mathcal{R}_k) := \sup_{u, v \in \mathcal{R}_k} \|u - v\|_2 \xrightarrow{k \rightarrow \infty} 0$$

Convergent collection of safe regions is useful because it guarantees to find the exact support of x^* . Before to construct a collection of a dynamic safe regions, we will state some insight on the primal-dual problem.

3.1 Gap

Definition 7 (Duality gap). *Let P and D be the objective functions of primal and dual problems, respectively. For any primal-dual pair $(x, u) \in F_P \times F_D$, where F_P and F_D denote the feasible sets of the primal and dual problems, we define the duality gap as*

$$\mathcal{G}(x, u) := P(x) - D(u)$$

.

Remark 4. *If P and D are continuous, therefore $\mathcal{G}$ is continuous and strictly decreases along convergent iterates.*

Proposition 6 (Weak duality). *Under the same conditions as Definition 7,*

$$\forall (x, u) \in F_P \times F_D, \quad \mathcal{G}(x, u) \geq 0$$

Proof. Consider the primal problem defined for $n > 0$, its objective function $P : \mathbb{R}^n \rightarrow \mathbb{R}$ and its constraints for $p, q \in \mathbb{N}$ such that

$$g_i(x) \leq 0 \quad \forall i \in \llbracket 1, p \rrbracket \quad \text{and} \quad h_j(x) = 0 \quad \forall j \in \llbracket 1, q \rrbracket$$

so we get the primal feasible set $F_P := \{x \in \mathbb{R}^n, \quad \forall i, g_i(x) \leq 0 \quad \text{and} \quad \forall j, h_j(x) = 0\}$

We define the Lagrangian, for $x \in \mathbb{R}^n$ and $u = (\lambda, \nu) \in \mathbb{R}^{+p} \times \mathbb{R}^q$,

$$\mathcal{L}(x, u) := P(x) + \sum_{i=1}^p \lambda_i g_i(x) + \sum_{j=1}^q \nu_j h_j(x)$$

We define the dual feasible set $F_D := \{u \in \mathbb{R}^{+p} \times \mathbb{R}^q, \inf_x \mathcal{L}(x, u) \neq -\infty\}$. The dual function associated to this primal problem is,

$$D : u \in F_D \mapsto \inf_x \mathcal{L}(x, u) \in \mathbb{R}$$

If $(x, u) \in F_P \times F_D$ then $\forall (i, j) \in \llbracket 1, p \rrbracket \times \llbracket 1, q \rrbracket$ we have $g_i(x) \leq 0$ and $h_j(x) = 0$, so,

$$L(x, u) \leq P(x)$$

Hence

$$D(u) \leq L(x, u) \leq P(x) \implies \mathcal{G}(x, u) = P(x) - D(u) \geq 0$$

□

Proposition 7 (Strong duality). *Under Slater's condition (Definition 2),*

$$\inf_{x \in F_P} P(x) = \sup_{u \in F_D} D(u)$$

Thus, it exists $x^* \in F_P$ and $u^* \in F_D$ such as,

$$\mathcal{G}(x^*, u^*) = 0$$

Proof. Firstly, by weak duality (Proposition 6),

$$\sup_{u \in F_D} D(u) \leq \inf_{x \in F_P} P(x)$$

Secondly, under Slater's condition, according to Theorem 1, there exist $x^* \in F_P$ and $u^* \in F_D$ verifying in particular, stationarity

$$0 \in \partial_x L(x^*, u^*) \implies D(u^*) = \inf_x L(x, u^*) = L(x^*, u^*)$$

And from complementary slackness and primal feasibility,

$$\forall (i, j) \in \llbracket 1, p \rrbracket \times \llbracket 1, q \rrbracket, \quad \lambda_i^* g_i(x^*) = 0, \quad h_j(x^*) = 0 \implies L(x^*, u^*) = P(x^*)$$

Therefore

$$\sup_u D(u) \geq D(u^*) = L(x^*, u^*) = P(x^*) \geq \inf_x P(x)$$

Finally, we get,

$$\sup_{u \in F_D} D(u) = \inf_{x \in F_P} P(x) \quad \text{and} \quad \mathcal{G}(x^*, u^*) = 0$$

□

The gap function measures how far the primal and dual objectives are from each other. It reflects the idea that the primal value is always greater than or equal to the dual one. In our LASSO setting, reducing the gap helps us track how close we are to the optimal solution. When strong duality holds, the gap becomes zero at the solution, meaning both primal and dual match. So the gap is useful as a stopping criterion as described below.

As mentioned earlier, in practice, LASSO solvers generate a sequence $(x_k)_{k \geq 0}$ that converge monotonically toward the optimal solution x^* . We can then define the corresponding dual sequence for our LASSO problem by $u_k := y - Ax_k$, which converges to u^* . By continuity of the gap function, we have

$$\mathcal{G}(x_k, u_k) \xrightarrow[k \rightarrow +\infty]{} \mathcal{G}(x^*, u^*) = 0$$

Each iterate thus gets closer to the optimal pair, and the duality gap provides an indication of how far we are from the true solution.

Lemma 1. *Let $\epsilon > 0$, x^* the optimal point of P and $(x, u) \in F_P \times F_D$,*

$$G(x, u) \leq \epsilon \implies P(x) - P(x^*) \leq \epsilon$$

Proof. Let $(x, u) \in F_P \times F_D$,

$$P(x) - P(x^*) = P(x) - D(u) + D(u) - P(x^*) \leq P(x) - D(u) = G(x, u)$$

since $D(u) \leq P(x^*)$ by weak duality. Hence if $G(x, u) \leq \epsilon$, then,

$$P(x) \leq P(x^*) + G(x, u) \iff P(x) - P(x^*) \leq \epsilon$$

so x is an ϵ -optimal solution of P . □

3.2 Gap Sphere Regions ($\mathcal{B}_k$)

Theorem 4 (Gap-Sphere [FGS15]). *For $\lambda > 0$, let $(x_k)_{k \geq 0}$ be the primal iterates produced by a convergent solver minimizing monotonically the objective function P_λ . From 3, we choose a dual point at iteration k as the projection⁶ on the feasible set of $\mathcal{D}_\lambda$, $(u_k)_{k \geq 0} := (y - Ax_k)_{k \geq 0}$*

For each iteration $k \geq 0$, let

$$r_k := \sqrt{2\mathcal{G}_\lambda(x_k, u_k)}$$

Hence, $(\mathcal{B}_k(u_k, r_k))_{k \geq 0}$, the collection along each iteraton k of ball centered at u_k with radius r_k , is a convergent collection of set region of u^ .*

Proof. Since

$$D_\lambda(u) = \langle y, u \rangle - \frac{1}{2}\|u\|_2^2,$$

we have $\partial D_\lambda(u) = y - u$ and $\partial^2 D_\lambda(u) = -I$. By the second order Taylor expansion around u^* , there exists ξ on $[u^*, u_k]$ such that

$$D_\lambda(u_k) = D_\lambda(u^*) + \partial D_\lambda(u^*)^\top (u_k - u^*) + \frac{1}{2} (u_k - u^*)^\top \partial^2 D_\lambda(\xi) (u_k - u^*),$$

hence using $\partial^2 D_\lambda(\xi) = -I$,

$$D_\lambda(u_k) = D_\lambda(u^*) + \partial D_\lambda(u^*)^\top (u_k - u^*) - \frac{1}{2}\|u_k - u^*\|_2^2.$$

Because u^* maximizes D_λ over the convex feasible set $\mathcal{F}_D$, its gradient satisfies $\partial D_\lambda(u^*)^\top (v - u^*) \leq 0$ for all feasible v . In particular, $\partial D_\lambda(u^*)^\top (u_k - u^*) \leq 0$. Therefore

$$D_\lambda(u_k) \leq D_\lambda(u^*) - \frac{1}{2}\|u_k - u^*\|_2^2 \implies D_\lambda(u^*) - D_\lambda(u_k) \geq \frac{1}{2}\|u_k - u^*\|_2^2.$$

Strong duality gives $D_\lambda(u^*) = P_\lambda(x^*)$ and $P_\lambda(x_k) \geq D_\lambda(u_k)$, so

$$\mathcal{G}_\lambda(x_k, u_k) = P_\lambda(x_k) - D_\lambda(u_k) \geq D_\lambda(u^*) - D_\lambda(u_k)$$

⁶Indeed, the dual problem $(\mathcal{D}_\lambda)$ in (4) can be rewritten as

$$(\mathcal{D}_\lambda): \min_{u \in \mathbb{R}^m} \|y - u\|_2 \quad \text{s.t.} \quad \|A^\top u\|_\infty \leq \lambda.$$

Hence u is the Euclidean projection of y onto the feasible set $\mathcal{F}_D = \{u : \|A^\top u\|_\infty \leq \lambda\}$.

Combining with the previous bound yields

$$\frac{1}{2} \|u_k - u^*\|_2^2 \leq \mathcal{G}_\lambda(x_k, u_k) \implies \|u_k - u^*\|_2 \leq \sqrt{2\mathcal{G}_\lambda(x_k, u_k)}$$

Defining $r_k = \sqrt{2\mathcal{G}_\lambda(x_k, u_k)}$ and using $u_k = y - Ax_k$, we obtain $\|u_k - u^*\|_2 \leq r_k$. Hence $\mathcal{B}_k = \{u : \|u - u_k\| \leq r_k\}$ is a *safe region* containing u^* .

Moreover, $(\mathcal{B}_k)_k$ converge because by continuity $\mathcal{G}_\lambda(x_k, u_k) \xrightarrow[k \rightarrow \infty]{} \mathcal{G}_\lambda(x^*, u^*) = 0$ so,

$$\text{diam}(\mathcal{B}_k) = 2r_k \xrightarrow[k \rightarrow \infty]{} 0$$

□

To improve solvers, the screening has to be effective as soon as possible and even if we have convergence it may be too slow in practice if the solver might have already identified x^* . We need to converge faster, aiming for *finite-time identification*.

Theorem 5 (Finite-time identification, [FGS15]). *Let $(\mathcal{R}_k)_{k \geq 0}$ be a convergent sequence of safe regions. Then there exists k_{safe} such that for all $k \geq k_{\text{safe}}$ the a set recovered from $(\mathcal{R}_k)$ coincides with the support of the optimal solution.*

Proof. Let x^* be the unique solution of the Lasso primal problem, and denote its support by

$$S := \text{supp}(x^*) = \{j : x_j^* \neq 0\}, \quad \bar{S} := \llbracket 1, n \rrbracket \setminus S$$

By the KKT conditions for $(\mathcal{P}_\lambda)$, the dual optimizer u^* satisfies

$$a_j^\top u^* = \begin{cases} \text{sign}(x_j^*) \lambda, & j \in S, \\ r_j, |r_j| < \lambda, & j \in \bar{S}. \end{cases}$$

Hence

$$|a_j^\top u^*| = \lambda \quad (\forall j \in S) \quad |a_j^\top u^*| < \lambda \quad (\forall j \in \bar{S})$$

We are given a sequence of safe regions $(\mathcal{R}_k)_{k \geq 0}$ such that

$$u^* \in \mathcal{R}_k \quad \text{for all } k, \quad \text{diam}(\mathcal{R}_k) \rightarrow 0 \quad \text{as } k \rightarrow \infty$$

That means, for $\epsilon > 0$, there exist K such that for all $k \geq K$, $\text{diam}(\mathcal{R}_k) < \epsilon$.

Define the recovered support at iteration k by

$$\hat{S}_k = \{j : \max_{u \in \mathcal{R}_k} |a_j^\top u| \geq \lambda\}$$

We will show that $\hat{S}_k = S$ for all large k .

(i) First of all, we show that inactive indices are excluded: Fix any $j \in \bar{S}$. Set $\delta_j = \lambda - |a_j^\top u^*| > 0$. Since $u \mapsto a_j^\top u$ is continuous and $\|a_j\|_2$ -Lipschitz and $\mathcal{R}_k$ converge towards $\{u^*\}$, there exists $k_j \in \mathbb{N}^*$ such that for $k \geq k_j$,

$$|a_j^\top u - a_j^\top u^*| \leq \|a_j\|_2 \|u - u^*\|_2 \leq \|a_j\|_2 \text{diam}(\mathcal{R}_k)$$

Hence, if we choose $\epsilon = \delta_j / \|a_j\|_2$, then for $k \geq k_j$,

$$\max_{u \in \mathcal{R}_k} |a_j^\top u - a_j^\top u^*| < \delta_j$$

Thus for $k \geq k_j$, we have $j \notin \hat{S}_k$. Indeed,

$$\max_{u \in \mathcal{R}_k} |a_j^\top u| \leq |a_j^\top u^*| + \max_{u \in \mathcal{R}_k} |a_j^\top u - a_j^\top u^*| < |a_j^\top u^*| + \delta_j = \lambda \implies \max_{u \in \mathcal{R}_k} |a_j^\top u| < \lambda$$

(ii) Now we show that active indices are never excluded: For any $j \in S$, since $u^* \in \mathcal{R}_k$ and $|a_j^\top u^*| = \lambda$, we have,

$$\max_{u \in \mathcal{R}_k} |a_j^\top u| \geq |a_j^\top u^*| = \lambda$$

so $j \in \widehat{S}_k$ for all k . Finally, Let $k_{\text{safe}} = \max_{j \in \bar{S}} k_j$, which is finite since $|\bar{S}| < \infty$. Then for all $k \geq k_{\text{safe}}$, part (i) gives $\widehat{S}_k \cap \bar{S} = \emptyset$ and part (ii) gives $S \subseteq \widehat{S}_k$. Hence

$$\widehat{S}_k = S \quad \forall k \geq k_{\text{safe}}$$

□

4 Numerical Experiments

We present two common algorithms for solving the LASSO problem. Then we will see how safe screening can be incorporated and accelerate them. We conclude this report with results on numerical experiments.

4.1 ISTA

Definition 8 (Proximity operator). *Let $\gamma > 0$ and convex function $g : \mathbb{R}^n \rightarrow \mathbb{R}$, for $n > 0$, we call proximal operator of g with γ ,*

$$\text{prox}_{\gamma, g} : v \in \mathbb{R}^n \mapsto \arg \min_{x \in \mathbb{R}^n} \left\{ g(x) + \frac{1}{2\gamma} \|x - v\|_2^2 \right\}$$

Iterative Soft-Thresholding Algorithm (ISTA) is a proximal gradient method, each iteration (w_k) performs a gradient descent step on the smooth loss followed by a soft-thresholding.

Definition 9 (ISTA, [DDM04]). *To minimize the problem $f + g$ with f convex, where its differential is L -Lipschitz and g convex. We approximate the minimizer w^* by $(w_k)_{k \geq 0}$ defined as, w_0 an initial point and with the update rule, for $\gamma > 0$,*

$$w^{(k+1)} = \text{prox}_{\gamma, g}(w_k - \gamma \partial f(w_k))$$

Remark 5 (Lasso explicit update). *For LASSO problem, to correspond with the precedent definition, we have,*

$$f : w \mapsto \frac{1}{2} \|Aw - b\|_2^2 \quad \text{and} \quad g : w \mapsto \lambda \|w\|_1$$

For $k \geq 0$, let $u_k = w_k - \gamma A^\top (Aw_k - b)$. Then, $\forall i \in \llbracket 1, n \rrbracket$,

$$w_{k+1}^{(i)} = \text{sign}(u_k^{(i)}) \max(|u_k^{(i)}| - \gamma\lambda, 0)$$

Proof. Since g is separable we have coordinate-wise on prox, hence, for $i \in \llbracket 1, n \rrbracket$,

$$w_{k+1}^{(i)} = \text{prox}_{\gamma, \lambda|\cdot|}(u_k^{(i)}) = \arg \min_{x \in \mathbb{R}} \left\{ \lambda|x| + \frac{1}{2\gamma} (x - u_k^{(i)})^2 \right\} \quad (\text{R1})$$

From Theorem 1, we denote x^* the minimizer of (R1) verifying,

$$0 \in \partial \left(x \mapsto \lambda|x| + \frac{1}{2\gamma} (x - u_k^{(i)})^2 \right) (x^*) = \lambda \partial|\cdot|_1(x^*) + \frac{1}{\gamma} (x^* - u_k^{(i)})$$

- $x^* > 0 \implies 0 = \lambda \cdot 1 + \frac{1}{\gamma} (x^* - u_k^{(i)}) \iff x^* = u_k^{(i)} - \gamma\lambda$ et $u_k^{(i)} > \gamma\lambda$
- $x^* < 0 \implies 0 = \lambda \cdot (-1) + \frac{1}{\gamma} (x^* - u_k^{(i)}) \iff x^* = u_k^{(i)} - \gamma\lambda$ et $u_k^{(i)} < -\gamma\lambda$
- $x^* = 0 \implies 0 \in \lambda[-1, 1] + \frac{1}{\gamma} (0 - u_k^{(i)}) \implies u_k^{(i)} \in [-\gamma\lambda, \gamma\lambda]$

Combining these three cases gives,

$$w_{k+1}^{(i)} = x^* = \begin{cases} u_k^{(i)} - \gamma\lambda, & u_k^{(i)} > \gamma\lambda \\ 0, & |u_k^{(i)}| \leq \gamma\lambda \\ u_k^{(i)} + \gamma\lambda, & u_k^{(i)} < -\gamma\lambda \end{cases} = \text{sign}(u_k^{(i)}) \max\{|u_k^{(i)}| - \gamma\lambda, 0\}$$

□

Usually we take $\gamma = \frac{1}{L}$ to exploit the regularity of f and get convergence of order 1.

Proposition 8. *Under the hypotheses of Definition 9, let $\gamma \in]0, \frac{2}{L}[$ and we denote $F := f + g$. The algorithm converge linearly,*

$$F(w_k) - F(w^*) = \mathcal{O}\left(\frac{1}{k}\right) \quad \text{as } k \rightarrow +\infty$$

Proof. Since ∂f is L -Lipschitz and $\gamma < \frac{2}{L}$, for $k \geq 0$,

$$f(w_{k+1}) \leq f(w_k) + \partial f(w_k)^\top (w_{k+1} - w_k) + \frac{1}{2\gamma} \|w_{k+1} - w_k\|^2$$

Adding $g(w_{k+1})$,

$$F(w_{k+1}) = f(w_{k+1}) + g(w_{k+1}) \leq f(w_k) + \text{prox}_{\gamma, g}(w_k) \leq f(w_k) + g(w_k) = F(w_k)$$

Thus $(F(w_k))_{k \geq 0}$ is non increasing. Next, we remark optimality condition at w_{k+1} gives,

$$0 \in \partial g(w_{k+1}) + \frac{1}{\gamma} (w_{k+1} - w_k + \gamma \partial f(w_k)) \iff \frac{w_k - w_{k+1}}{\gamma} - \partial f(w_k) \in \partial g(w_{k+1})$$

Then, we set,

$$s_{k+1} := \frac{w_k - w_{k+1}}{\gamma} + (\partial f(w_{k+1}) - \partial f(w_k)) \in \partial F(w_{k+1})$$

And because F is convex, we have in particular,

$$F(w^*) \geq F(w_{k+1}) + s_{k+1}^\top (w^* - w_{k+1}) \iff F(w_{k+1}) - F(w^*) \leq \frac{1}{\gamma} (w_k - w_{k+1})^\top (w_{k+1} - w^*)$$

Using the three-point identity ⁷ $(w_k - w_{k+1})^\top (w_{k+1} - w^*) = \frac{1}{2} (\|w_k - w^*\|^2 - \|w_{k+1} - w^*\|^2 - \|w_k - w_{k+1}\|^2)$.
Hence

$$F(w_{k+1}) - F(w^*) \leq \frac{1}{2\gamma} (\|w_k - w^*\|^2 - \|w_{k+1} - w^*\|^2)$$

By non increasing and then by telescoping,

$$k(F(w_k) - F(w^*)) \leq \sum_{i=0}^{k-1} F(w_{i+1}) - F(w^*) \leq \frac{\|w_0 - w^*\|^2}{2\gamma}$$

Hence, as $k \rightarrow \infty$,

$$F(w_k) - F(w^*) = \mathcal{O}\left(\frac{1}{k}\right)$$

□

We write the algorithm used in following experiments.

Algorithm 1 ISTA for LASSO

Require: $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $\lambda > 0$, $\varepsilon > 0$

1: $L \leftarrow \|A\|_2^2$, $\gamma \leftarrow 1/L$, $w \leftarrow 0$

▷ Initialization on primal problem

2: **repeat**

3: $u \leftarrow w - \gamma A^\top (Aw - b)$

▷ gradient step on $f : w \mapsto \frac{1}{2} \|Aw - b\|_2^2$

4: $w \leftarrow \text{sign}(u) \odot \max(|u| - \gamma, \lambda, 0)$

▷ $\odot$ = element-wise $\text{prox}_{\gamma, \lambda \|\cdot\|_1}$

5: **until** $\|A^\top (Aw - b)\|_\infty \leq \lambda + \varepsilon$

6: **return** w

⁷ $\forall a, b, c \in \mathbb{R}^n$, $2\langle a - b, b - c \rangle = \|a - c\|^2 - \|a - b\|^2 - \|b - c\|^2$

4.2 Coordinate Descent

Coordinate Descent (CD) solves by successively minimizing along one coordinate while keeping the others fixed.

Definition 10 (Cyclic coordinate descent). *Let $n > 0$, f convex, L -Lipschitz, g convex and separable⁸. We denote w^* the minimizer of $F : w \in \mathbb{R}^n \mapsto f(w) + g(w)$*

Set w_0 an initial point, we approximate w^ with the sequence (w_k) in updating each component at one time. For $k > 0$, and $\forall i \in \llbracket 1, n \rrbracket$*

$$w_{k+1}^{(i)} := \arg \min_{t \in \mathbb{R}} f(w_k + (t - w_k^{(i)})e^{(i)}) + g_i(t)$$

Theorem 6 (Explicit coordinate-wise update for Lasso). *Consider the Lasso problem, let $a_j \in \mathbb{R}^m$ denote the j -th column of A , and let $r = b - Aw$ be the current residual. If all coordinates $w^{(k)}$ for $k \neq j$ are held fixed, then the exact minimizer with respect to the single coordinate $w^{(j)}$ is given by*

$$z_j := a_j^\top r + \alpha_j w^{(j)} \quad w_{(new)}^{(j)} = \frac{1}{\|a_j\|_2^2} \text{sign}(z_j) \max\{|z_j| - \lambda, 0\} \quad r_{(new)} = r - (w_{(new)}^{(j)} - w^{(j)}) a_j$$

Proof. Fix a coordinate index j . At iteration t , write $w_t = (w_t^{(1)}, \dots, w_t^{(n)})$ and denote the current residual

$$r_t = b - Aw_t$$

We update only the j -th component $w_t^{(j)}$ by solving

$$\min_{u \in \mathbb{R}} \frac{1}{2} \|A(w_t + (u - w_t^{(j)})e_j) - b\|_2^2 + \lambda |u|$$

since the other $|w_t^{(k)}|$ ($k \neq j$) remain constant. Noting that

$$A(w_t + (u - w_t^{(j)})e^{(j)}) = Aw_t + (u - w_t^{(j)})a_j = b - r_t + (u - w_t^{(j)})a_j$$

So we have,

$$\|A(w_t + (u - w_t^{(j)})e^{(j)}) - b\|_2^2 = \|-r_t + (u - w_t^{(j)})a_j\|_2^2 = \|r_t\|_2^2 - 2(u - w_t^{(j)})(a_j^\top r_t) + (u - w_t^{(j)})^2 \|a_j\|_2^2$$

Because $\|r_t\|_2^2$ is constant in u , the subproblem becomes

$$\min_u \left\{ \frac{\|a_j\|_2^2}{2} u^2 - (a_j^\top r_t + \|a_j\|_2^2 w_t^{(j)}) u + \lambda |u| \right\}$$

Set $\alpha_j := \|a_j\|_2^2$ and $z_j := a_j^\top r_t + \alpha_j w_t^{(j)}$. Thus we minimize

$$\varphi(u) = \frac{\alpha_j}{2} u^2 - z_j u + \lambda |u|$$

Because φ is strictly convex and separable in sign of u , set u^* the unique minimizer of $\varphi(u)$, verifying,

$$0 \in \partial\varphi(u^*) = \frac{\alpha_j}{2} (u^*)^2 - z_j u^* + \lambda |u^*|$$

- $u^* > 0 \implies 0 = \alpha_j u^* - z_j + \lambda \iff u^* = \frac{z_j - \lambda}{\alpha_j}$ et $z_j > \lambda$
- $u^* < 0 \implies 0 = \alpha_j u^* - z_j - \lambda \iff u^* = \frac{z_j + \lambda}{\alpha_j}$ et $z_j < -\lambda$
- $u^* = 0 \implies 0 \in -z_j + \lambda[-1, 1] \iff z_j \in [-\lambda, \lambda]$

⁸ $\exists (g_i)_{1 \leq i \leq n}$ such that $g(w) = \sum_{i=1}^n g_i(w^{(i)})$

By combining these three, we get,

$$w_{t+1}^{(j)} = u^* = \frac{1}{\|a_j\|_2^2} \text{sign}(z_j) \max\{|z_j| - \lambda, 0\}$$

Hence

$$w_{t+1}^{(j)} = \frac{1}{\alpha_j} \text{sign}(z_j) \max\{|z_j| - \lambda, 0\}, \quad w_{t+1}^{(k)} = w_t^{(k)} \quad (\forall k \neq j)$$

Finally, writing $\Delta = w_{t+1}^{(j)} - w_t^{(j)}$, the residual updates by

$$r_{t+1} = b - A w_{t+1} = b - (A w_t + \Delta a_j) = r_t - \Delta a_j$$

□

Remark 6. The updated residual can be maintained in $O(m)$ time. Indeed, It takes $2m$ arithmetic operations to evaluate r_{new} from r .

Proposition 9 (Convergence rate, [Tse01; FHT10]). Assume A contains no null column⁹ and that the Lasso solution w^* is unique. For the cyclic CD sequence (w_k) defined in Definition 10,

$$F(w_k) - F(w^*) = \mathcal{O}\left(\frac{1}{k}\right)$$

Proof. For LASSO corresponding to Definition 10 let

$$f(w) = \frac{1}{2} \|A w - b\|_2^2, \quad g(w) = \lambda \|w\|_1.$$

Because each column a_j of A is nonzero, the one dimensional subproblem in coordinate j is strongly convex with modulus $\alpha_j = \partial^2 F_j = \|a_j\|_2^2 > 0$. The exact minimizer over $u \mapsto F(w_k + (u - w_k^{(j)} e^{(j)}))$ is

$$w_{k+1}^{(j)} = \arg \min_{u \in \mathbb{R}} \left\{ f(w_k + (u - w_k^{(j)} e^{(j)})) + \lambda |u| \right\}, \quad w_{k+1}^{(i)} = w_k^{(i)} \quad \forall i \neq j$$

The strong-convexity implies

$$f(w_k + (u - w_k^{(j)} e^{(j)})) + \lambda |u| - \left[f(w_k) + \lambda |w_k^{(j)}| \right] \geq \frac{\alpha_j}{2} (u - w_k^{(j)})^2 + \left\langle \partial_j f(w_k), u - w_k^{(j)} \right\rangle - \lambda (|w_k^{(j)}| - |u|)$$

Hence,

$$F(w_k) - F(w_{k+1}) = F(w_k) - \min_u \left\{ f(w_k + (u - w_k^{(j)} e^{(j)})) + \lambda |u| \right\} \geq \frac{\alpha_j}{2} (w_k^{(j)} - w_{k+1}^{(j)})^2$$

Let w_k the iterate at the beginning of cycle k . A *cycle* consists of updating each coordinate $1, 2, \dots, n$ exactly once in order. Formally:

$$w_{k,0} = w_k,$$

$$\text{for } \ell = 1, 2, \dots, n: \quad w_{k,\ell}^{(i)} = \begin{cases} \arg \min_{u \in \mathbb{R}} F(w_{k,\ell-1} + (u - w_{k,\ell-1}^{(\ell)} e_\ell)) & i = \ell \\ w_{k,\ell-1}^{(i)} & i \neq \ell \end{cases}$$

$$w_{k+1} = w_{k,n}$$

Since updating coordinate ℓ satisfies

$$F(w_{k,\ell-1}) - F(w_{k,\ell}) \geq \frac{\|a_\ell\|_2^2}{2} (w_{k,\ell-1}^{(\ell)} - w_{k,\ell}^{(\ell)})^2$$

⁹A zero column never influences the objective and can be discarded in a preprocessing step

summing $\ell = 1$ to n gives

$$F(w_{k,0}) - F(w_{k,n}) = \sum_{\ell=1}^n [F(w_{k,\ell-1}) - F(w_{k,\ell})] \geq \sum_{\ell=1}^n \frac{\|a_\ell\|_2^2}{2} (w_{k,\ell-1}^{(l)} - w_{k,\ell}^{(l)})^2$$

But $w_{k,n} = w_{k+1}$, and $\sum_{\ell=1}^n (w_{k,\ell-1}^{(l)} - w_{k,\ell}^{(l)})^2 = \|w_k - w_{k+1}\|_2^2$. Setting $\rho = \min_{1 \leq j \leq n} \|a_j\|_2^2 > 0$ yields

$$F(w_k) - F(w_{k+1}) \geq \sum_{\ell=1}^n \frac{\|a_\ell\|_2^2}{2} (w_{k,\ell-1}^{(l)} - w_{k,\ell}^{(l)})^2 \geq \frac{\rho}{2} \|w_k - w_{k+1}\|_2^2$$

By telescoping

$$\sum_{t=0}^{K-1} [F(w_t) - F(w_{t+1})] = F(w_0) - F(w_K) \leq F(w_0) - F(w^*) < \infty$$

and hence $\sum_{t=0}^{\infty} \|w_t - w_{t+1}\|_2^2 < \infty$, implying $\|w_t - w_{t+1}\|_2 \rightarrow 0$.

Let $\Delta_k = F(w_k) - F(w^*)$. By convexity and uniqueness of w^* , there is $C > 0$ ([Tse01]) such that

$$F(w_k) - F(w^*) \leq C \|w_k - w_{k+1}\|_2$$

Combining these two inequalities:

$$\Delta_k - \Delta_{k+1} = F(w_k) - F(w_{k+1}) \geq \frac{\rho}{2} \|w_k - w_{k+1}\|_2^2 \geq \frac{\rho}{2} \left(\frac{\Delta_k}{C}\right)^2 = \mu \Delta_k^2, \quad \mu = \frac{\rho}{2C^2}$$

Rearrange:

$$\Delta_{k+1} \leq \Delta_k - \mu \Delta_k^2 = \Delta_k (1 - \mu \Delta_k)$$

Since $\Delta_k \rightarrow 0$, for large k we have $\mu \Delta_k < 1$, so

$$\frac{1}{\Delta_{k+1}} \geq \frac{1}{\Delta_k} \frac{1}{1 - \mu \Delta_k} \geq \frac{1}{\Delta_k} + \mu$$

By induction, $\frac{1}{\Delta_k} \geq \frac{1}{\Delta_0} + \mu k$, hence $\Delta_k \leq (\frac{1}{\Delta_0} + \mu k)^{-1} = O(\frac{1}{k})$. Consequently,

$$F(w^k) - F(w^*) = O\left(\frac{1}{k}\right)$$

□

Remark 7 (Computational cost). A complete cycle updates every coordinate once and requires $\mathcal{O}(nm)$ operations if all inner products are recomputed. After several iterations many coordinates become exactly zero. Let $q \leq n$ denote the number of indices with $w_k^{(j)} \neq 0$. Caching $a_j^\top r_k$ for inactive j reduces the effective cost of a cycle to $\mathcal{O}(mq)$

Algorithm 2 Coordinate Descent for LASSO

Require: $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $\lambda > 0$, $\varepsilon > 0$

```

1:  $w \leftarrow 0$ ,  $r \leftarrow b$ 
2: repeat
3:   for  $j = 1$  to  $n$  do
4:      $z \leftarrow a_j^\top r + \|a_j\|_2^2 w^{(j)}$ 
5:      $w_{\text{new}}^{(j)} \leftarrow \text{sign}(z), \max(|z| - \lambda, 0) / \|a_j\|_2^2$ 
6:      $r \leftarrow r + a_j (w^{(j)} - w_{\text{new}}^{(j)})$ 
7:      $w^{(j)} \leftarrow w_{\text{new}}^{(j)}$ 
8:   end for
9: until  $|A^\top r|_\infty \leq \lambda + \varepsilon$ 
10: return  $w$ 
```

4.3 Implementing the Gap Safe Sphere Rule

In the Section 2 we developed a safe rule to discard features unnecessary during the optimization process. Then, in Section 3 we enhanced this rule with *Gap Safe Sphere*, which is dynamic, enabling feature screening throughout the iterative solving process.

Practically, the Gap Safe Sphere rule is not applied at every iteration due to its computational overhead. Moreover, since the radius of the Gap Sphere, determined by the duality gap, typically shrinks slowly from one iteration to the next, frequent screening often yields minimal incremental gain. Therefore, we apply this rule periodically, every $freq_screen$ iterations. Next, we activate it only after the duality gap shrinks below a fraction $\rho \in]0, 1[$ of its initial value, thereby guaranteeing that the solver has sufficiently approached the optimal region.

As already said, we can implement safe rule on any monotone iterative solver. Algorithmically, we implement in our particular case, Gap Safe Sphere rule, converging to an ϵ -optimal solution according to Lemma 1,

Algorithm 3 Solver with Gap Safe Sphere Screening for LASSO

Require: $A \in \mathbb{R}^{m \times n}$, $y \in \mathbb{R}^m$, $\lambda > 0$, $\varepsilon > 0$, $\rho \in]0, 1[$, $freq_screen \in \mathbb{N}^*$

```

1:  $iter \leftarrow 0$ 
2:  $x \leftarrow 0$                                 ▷ Initialize primal solution
3:  $u \leftarrow y - Ax$                             ▷ Compute initial dual solution
4:  $G_0 \leftarrow P_\lambda(x) - D_\lambda(u)$                 ▷ Compute initial duality gap
5: Set  $G \leftarrow G_0$ 
6: Set active feature set  $\mathcal{A} \leftarrow \llbracket 1, n \rrbracket$ 
7: while  $G > \varepsilon$  do
8:    $iter \leftarrow iter + 1$ 
9:   Perform one solver iteration to update primal solution  $x$ 
10:   $u \leftarrow y - Ax$                             ▷ Update dual solution
11:   $G \leftarrow P_\lambda(x) - D_\lambda(u)$                 ▷ Update duality gap
12:  if  $G \leq \rho G_0$  and  $iter \equiv 0[ freq\_screen]$  then
13:     $R \leftarrow \sqrt{2G}$ 
14:    for  $j$  in active feature indices  $\mathcal{A}$  do
15:      if  $|a_j^T u| + R \|a_j\|_2 < \lambda$  then                ▷ Gap Safe Sphere rule
16:        Remove  $j$  from active set  $\mathcal{A}$ 
17:      end if
18:    end for
19:  end if
20: end while
21: return  $x$ 

```

We conducted experiments with $n = 10^4$ features fixed. The design matrices A are Gaussian, with the number of observations m exploring performance in regimes where $m \ll n$. The parameter λ was selected as a fraction of $\lambda_{\max} = \|A^T y\|_\infty$. The solutions in these problems are sparse. In compressed sensing it's assume problem in high dimension has sparse solution. Hence, the Gap Safe rule can discard irrelevant features more aggressively.

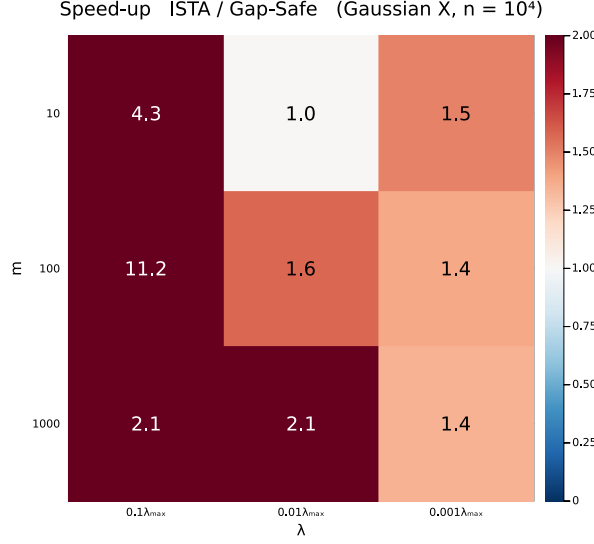


Figure 3: Performance heatmap (ISTA vs. Gap Safe ISTA) on Gaussian matrices. The numerical values correspond to the ISTA execution time divided by Gap Safe execution time.

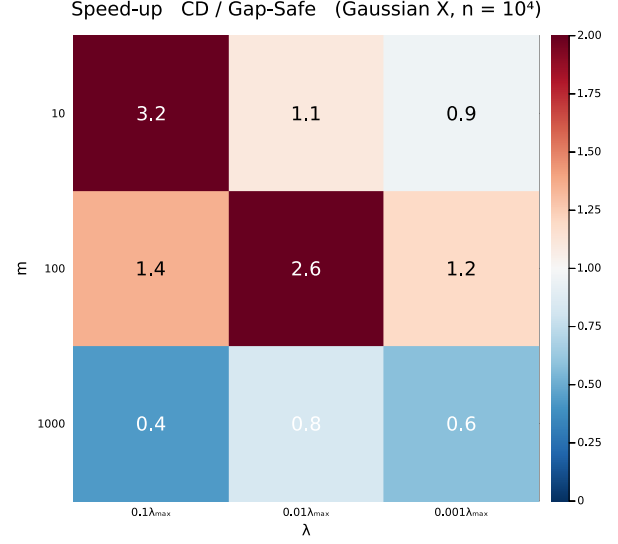


Figure 4: Performance heatmap (Coordinate Descent vs. Gap Safe CD) on Gaussian matrices. The numerical values correspond to the Coordinate Descent execution time divided by Gap Safe execution time.

On Figure 3 and Figure 4, higher (redder) values indicate greater speed-up using Gap Safe Sphere rule. Results, averaged over two runs. We see significant performance improvements (value are greater than 1) on the left figure. However, surprisingly, Coordinate Descent does not exhibit consistent improvements across all tested configurations. Due to time constraints, we could not further analyze these outcomes to clarify the underlying causes of these inconsistent results.

Further investigation is required to analyze performance when $m \approx n$ or when λ approaches zero, scenarios that represent limits of the Gap Safe method’s efficacy. Additionally, it could be interesting to examining sparse matrices or dense support solutions and too different design matrices. Indeed, Coordinate Descent, due to its dimension-wise updates, should theoretically benefit significantly from screening. The question to known on what configuration with what hyper-parameters (ϵ , *freq.screen*) is not well understand by authors but the theory is interesting and some papers encountered positives results, see [GVR10] [FGS15] [Tra+22]

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